Embedded Controller Quiz Review

Review on SysTick, Interrupt, UART

Concept: SysTick

- Explain how SysTick works in terms of its components (e.g., the down-counter).
- Explain how to configure the Tick period, including how to calculate the reload value.

Concept: Interrupt

- Explain the difference between polling and interrupts.
- Define the following terms: External Interrupt, Exception, ISR, IRQn, and NVIC.
- Explain how an interrupt is managed by the system.
- Explain the process that occurs when multiple ISRs with different priorities are triggered.

Concept: UART

- What is serial communication? Explain how 8-bit data is transmitted and received.
- Define odd/even parity and baud rate.
- What are the major differences between RS-232 and RS-485?
- Explain in detail when the TXE and RXNE flags become 'H' (High) or 'L' (Low). (Use figures to support your explanation).
- What is SPI, and how does it differ from UART?

Register Configuration

- Able to configure specific peripheral registers based on given descriptions or instructions using the reference manual.

```
void SysTick_init(void)

void EXTI_init(PinName_t pinName, uint32_t trig_type, uint32_t priority);

bool is_pending_EXTI(PinName_t pinName);

void USART_baud(USART_TypeDef* USARTx, uint32_t baud);
```